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B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, JUNE 2025
Third Semester
21CSC201J - DATA STRUCTURES AND ALGORITHMS
(For the candidates admitted from the academic year 2021-2022 TO 2024-2025)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer ALL Questions

Marks BL CO PO

- | | | | | | |
|----|---|---|---|---|---|
| 1. | How do you access a member of a structure using a pointer in C?
A) ptr.member B) ptr->member
C) *ptr.member D) ptr&.member | 1 | 1 | 1 | 2 |
| 2. | What is the output of the following code?
<pre>int x = 10; int *p = &x; printf("%d", *p);</pre> A) Address of x B) 10
C) Garbage value D) Compilation error | 1 | 2 | 1 | 1 |
| 3. | Which of the following statements about matrix multiplication is true?
A) Matrix multiplication is commutative B) Matrix multiplication is associative
C) Matrix multiplication is distributive D) Both b and c over addition | 1 | 2 | 1 | 1 |
| 4. | What is the best time complexity of a linear search?
A) O(1) B) O(n)
C) O(log n) D) O(n ²) | 1 | 1 | 1 | 2 |
| 5. | In a singly linked list, how is an element inserted at the beginning of the list?
A) By creating a new node and updating the head pointer B) By shifting all elements one position to the right
C) By swapping the first two nodes D) By appending the element at the end | 1 | 3 | 2 | 2 |
| 6. | What is the disadvantage of inserting an element at the beginning in an array-based implementation of List ADT?
A) Requires extra space B) Takes constant time
C) Requires shifting all elements D) Cannot be done | 1 | 2 | 2 | 3 |
| 7. | How is the end of the list detected in a circular singly linked list?
A) When the next pointer is NULL B) When the next pointer points to the head
C) When the previous pointer is NULL D) When the previous pointer points to the tail | 1 | 2 | 2 | 2 |

8. In the Josephus problem, how is the next person to be eliminated chosen in a circular linked list? 1 3 2 3
 A) Randomly B) By a fixed number of steps
 C) By a linear search D) By sorting the list
9. What is the time complexity of the push operation in a stack implemented using an array? 1 1 3 2
 A) $O(1)$ B) $O(n)$
 C) $O(\log n)$ D) $O(n \log n)$
10. What is the postfix expression for the infix expression $(A + B) * (C + D)$? 1 3 3 2
 A) $AB + CD + *$ B) $AB + C + D *$
 C) $AB + CD * +$ D) $ABCD + * +$
11. How many moves are required to solve the Tower of Hanoi problem with 4 disks? 1 3 3 2
 A) 16 B) 15
 C) 14 D) 24
12. In a circular queue implemented using an array, how do you check if the queue is full? 1 2 3 3
 A) $front == rear$ B) $(rear + 1) \% size == front$
 C) $front == size$ D) $rear == size$
13. In a binary tree, the maximum number of nodes at level L is: 1 1 4 3
 A) 2^L B) $2^{(L-1)}$
 C) $2^{(L+1)}$ D) L^2
14. A complete binary tree is defined as: 1 1 4 2
 A) A binary tree in which all levels are completely filled. B) A binary tree in which all levels except possibly the last are completely filled.
 C) A binary tree with nodes having at most two children. D) A binary tree in which each node has exactly two children.
15. In an AVL tree, which rotation is applied when the tree becomes left-heavy due to the insertion of a new node in the left subtree of the left child? 1 3 4 2
 A) Right rotation B) Left rotation
 C) Left-right rotation D) Right-left rotation
16. What is the maximum height of a Red-Black tree with n nodes? 1 2 4 3
 A) $2 * \log(n+1)$ B) $2 * \log(n)$
 C) $\log(n+1)$ D) $\log(n)$
17. Quadratic probing is primarily used to: 1 2 5 3
 A) Avoid clustering. B) Handle large datasets.
 C) It is only applicable to undirected graphs. D) All edges must point in the same direction.
18. In a topological sorting of a graph, which of the following is true? 1 2 5 3
 A) The graph must be cyclic. B) It is possible to have multiple valid orderings.
 C) It is only applicable to undirected graphs. D) All edges must point in the same direction.
19. Which of the following data structures is commonly used to implement Kruskal's algorithm? 1 1 5 2
 A) Stack B) Queue
 C) Disjoint Set (Union-Find) D) Priority Queue
20. In Dijkstra's algorithm, which data structure is commonly used to select the next vertex with the minimum distance? 1 1 5 3
 A) Stack B) Queue
 C) Priority Queue (Min-Heap) D) Hash Table

PART - B (5 x 8 = 40 Marks)**Marks BL CO PO**Answer **ALL** the Questions

- 21 a. Define and explain the differences between primitive data types and structures in programming. Provide an example in C demonstrating the declaration and usage of both. 8 2 1 2
(OR)
b. Discuss dynamic memory allocation in C and illustrate how to dynamically allocate memory for a 2D matrix. Also, provide a simple example of matrix multiplication. 8 2 1 3
- 22 a. Define and differentiate between Singly Linked List, Doubly Linked List, and Circular Linked List. Provide a scenario where each type is preferred. 8 2 2 3
(OR)
b. Explain the Josephus Problem. Describe how a Circular Linked List can be used to solve this problem. Write the algorithm for the same. 8 2 2 2
- 23 a. Describe the priority queue and its implementation using a linked list. How does it differ from a normal queue? 8 2 3 3
(OR)
b. What is a circular queue? How is it different from a linear queue? Illustrate with an example. 8 2 3 3
- 24 a. Describe the different ways to represent a binary tree in memory. Compare the advantages and disadvantages of using a linked representation versus an array representation. 8 2 4 3
(OR)
b. Explain the properties of a Red-Black Tree. How do these properties ensure that the tree remains balanced? Illustrate with an example. 8 2 4 3
- 25 a. Describe Depth-First Search (DFS) and Breadth-First Search (BFS) graph traversal techniques. Compare their applications. 8 2 5 2
(OR)
b. Explain the process of topological sorting of a Directed Acyclic Graph (DAG). What are its applications? 8 2 5 3

PART - C (1 x 15 = 15 Marks)**Marks BL CO PO**Answer **ANY ONE** Questions

- 26 Discuss the importance of mathematical notations in analyzing algorithms. Explain asymptotic notations Big O, Omega, and Theta with examples, and describe how they are used to express the efficiency of algorithms. 15 2 1 3
- 27 Explain the concept of open addressing in hashing. Discuss the various probing techniques used in open addressing, including linear probing, quadratic probing, and double hashing. Evaluate their performance in terms of clustering and efficiency. 15 2 5 3
